

SeatIQ: Roster-Aware Seat Capacity Planning for Contact Centers

Bringing workforce management intelligence to a problem real-estate tools were never built to solve

Akanchha Agarwal • WFM Simplified • July 2026

Executive Summary

Contact centers face a seat planning paradox: they operate some of the most densely scheduled real estate in the corporate world, yet plan that space with tools designed for conventional offices. Commercial workplace platforms such as Tango, Robin, and Condeco treat a seat as a bookable asset with a binary state — reserved or free. A contact center seat is something different: a shared resource cycled across overlapping 24/7 shifts, governed by client segregation clauses, transport windows, and seat-sharing ratios that routinely place 1,300 rostered agents across 900 physical positions.

This paper proposes SeatIQ, a seat capacity planning tool that starts from the roster rather than the floor plan. SeatIQ computes seat demand at 30-minute interval granularity directly from scheduled shifts, formulates team-to-bay allocation as a constraint optimization problem, and lets planners simulate what-if scenarios — hiring batches, work-from-home policy shifts, shift redesigns — before a single desk is moved. The result is a planning capability that facilities tools cannot offer, because it is built on the discipline that actually drives contact center occupancy: workforce management.

1. The Problem: Seat Planning Without Shift Awareness

In a hybrid corporate office, seat planning is a demand-smoothing exercise — employees come and go, and utilization sensors or booking data reveal patterns over time. In a contact center, occupancy is not discovered; it is scheduled. The roster determines, weeks in advance and to the half-hour, exactly how many people will need a seat on every floor. Yet the tools used to manage that space ignore the roster entirely.

The consequences are familiar to anyone who has run capacity planning for a BPO delivery center:

- **Peak-hour collisions.** Overlapping shift boundaries (a 1:30 PM login wave arriving before the 2:00 PM logout wave leaves) create predictable 30–60 minute crunches that booking tools register only after agents are standing in aisles.
- **Guesswork seat-sharing ratios.** Ratios like 1.4 agents per seat are set annually as blunt averages, rather than derived per program from actual shift overlap mathematics — leaving some floors starved and others 40% empty at 3 AM.
- **Manual, fragile allocation.** Mapping programs to bays is done by drag-and-drop and institutional memory. Client contract constraints — dedicated floors, data segregation, voice/back-office separation — live in planners' heads, and every ramp or shift change triggers days of rework.
- **No forward view.** When a client confirms a 150-agent ramp, the question “do we have the seats, and where?” is answered with spreadsheets built from scratch each time.

Even the incumbent tools acknowledge the experience gap: user reviews of leading workplace apps cite dated interfaces and too many steps to find a seat. But the deeper failure is conceptual, not cosmetic. A real-estate platform optimizes square footage; a contact center needs to optimize the intersection of rosters and square footage. That intersection is a WFM problem, and no mainstream tool treats it as one.

2. The SeatIQ Approach

SeatIQ is organized as three connected modules, each answering one planning question.

Module 1 — Seat Demand Engine: How many seats, when?

The engine ingests the roster (agent, program, shift pattern, date, work mode, leave status) and computes concurrent seated headcount for every 30-minute interval, by program and floor. Work-from-home flags, training blocks, and leave are netted out so demand reflects bodies-in-building, not headcount-on-paper. Outputs include interval-level demand heatmaps (day × time), peak concurrency per program, and true seat-sharing ratios derived from overlap mathematics rather than assumption. This module alone replaces the fragile Excel models most centers depend on today.

Module 2 — Allocation Optimizer: Who sits where?

Given floor and bay capacities plus business rules, SeatIQ formulates seat allocation as a mixed-integer optimization problem. Constraints encode reality: programs cluster contiguously; client-dedicated or segregated areas are inviolable; voice teams stay clear of noise-sensitive zones; sharing ratios respect shift overlap limits. The solver proposes allocations that minimize fragmentation and constraint violations, and — critically — flags infeasibility explicitly, so a planner knows the difference between “tight” and “impossible” before committing to a client.

Module 3 — Scenario Planner: What happens if?

Planners adjust levers — add a 100-agent hiring batch, shift WFH from 30% to 50%, move a program’s window by an hour — and watch demand curves and allocations recompute live. A forecasting extension projects seat demand a quarter ahead from hiring plans and attrition assumptions, converting seat planning from a reactive scramble into a forward-looking capacity discipline, exactly as interval staffing forecasts did for scheduling.

How SeatIQ differs from booking-first platforms:

Dimension	Booking-first tools (e.g., Tango)	SeatIQ
Source of truth	Reservations and sensor data (backward-looking)	The roster (forward-looking, interval-level)
Shift awareness	None; a day is one occupancy state	Native; 30-minute concurrency with overlap math
Allocation method	Manual drag-and-drop	Constraint optimization with violation flags
Client constraints	Held in planners’ heads	Encoded as hard constraints in the model
Planning horizon	Today and this week	What-if scenarios and quarterly forecasts

3. Methodology and Technical Design

SeatIQ is deliberately built on a transparent, auditable analytics stack rather than a black box:

- **Demand computation.** Python (pandas) time-series aggregation transforms roster records into interval concurrency. Every number traces back to named shifts — planners can defend the output to operations and clients.
- **Optimization.** Team-to-bay assignment is modeled as a mixed-integer program (PuLP / Google OR-Tools), with capacities, adjacency, and segregation expressed as explicit constraints and fragmentation as the objective. Solver output includes binding constraints, so planners see why an allocation looks the way it does.
- **Interface.** A Streamlit web application: upload a roster CSV and floor configuration, explore heatmaps and allocations, adjust scenario sliders. No installation, no IT project — a working decision tool in the browser.

- **Data and privacy.** The reference implementation ships with a synthetic roster generator (1,200+ agents, four programs, realistic shift mixes and messiness), so the methodology can be demonstrated, validated, and extended without exposing any client or employee data.

4. Expected Outcomes

For a mid-sized delivery center (1,000–1,500 agents), the measurable value concentrates in four areas: recovery of stranded capacity by replacing blanket sharing ratios with per-program overlap-derived ratios, typically exposing 5–10% hidden seat headroom; elimination of peak-hour collisions through shift-boundary visibility; hours-to-minutes reduction in reallocation effort when rosters change; and credible, same-day answers to ramp feasibility questions during client solutioning. Beyond the numbers, SeatIQ repositions seat planning as an extension of capacity planning — owned by the discipline that already forecasts, schedules, and staffs the operation.

5. Implementation Roadmap

Phase	Scope	Deliverable
Phase 1	Synthetic roster generator; interval-level seat demand engine; heatmaps and sharing-ratio analytics	Validated demand model on 1,200-agent synthetic dataset
Phase 2	Constraint model for bays, segregation, adjacency; MIP solver integration; infeasibility diagnostics	Automated allocation proposals with violation flags
Phase 3	Streamlit application; scenario sliders (WFH %, hiring batches, shift moves)	Interactive planning tool, deployable to the browser
Phase 4	Quarterly seat demand forecasting from hiring and attrition plans	Forward capacity view integrated into scenario planner

Phases are sequenced so each stands alone: the demand engine delivers value before the optimizer exists, and the optimizer before the interface. Total build effort for the reference implementation is estimated at eight to ten weeks of part-time development.

6. Conclusion

The seat planning gap in contact centers is not a software gap; it is a framing gap. As long as seats are managed as real estate, planners will keep translating rosters into occupancy by hand, and tools will keep reporting yesterday's utilization instead of next month's crunch. SeatIQ closes that gap by making the roster the source of truth and treating allocation as the optimization problem it genuinely is. The techniques involved — interval analytics, constraint programming, scenario simulation — are proven and accessible; what has been missing is a tool that applies them with workforce management domain knowledge built in. This proposal outlines exactly that.

About the author: Akanchha Agarwal is a workforce management professional with five years of experience in forecasting, capacity planning, and operations analytics across global contact center programs, and the creator of WFM Simplified, an educational series on workforce management and analytics. She holds an M.Tech from IIT Kanpur and a Lean Six Sigma Black Belt certification.